

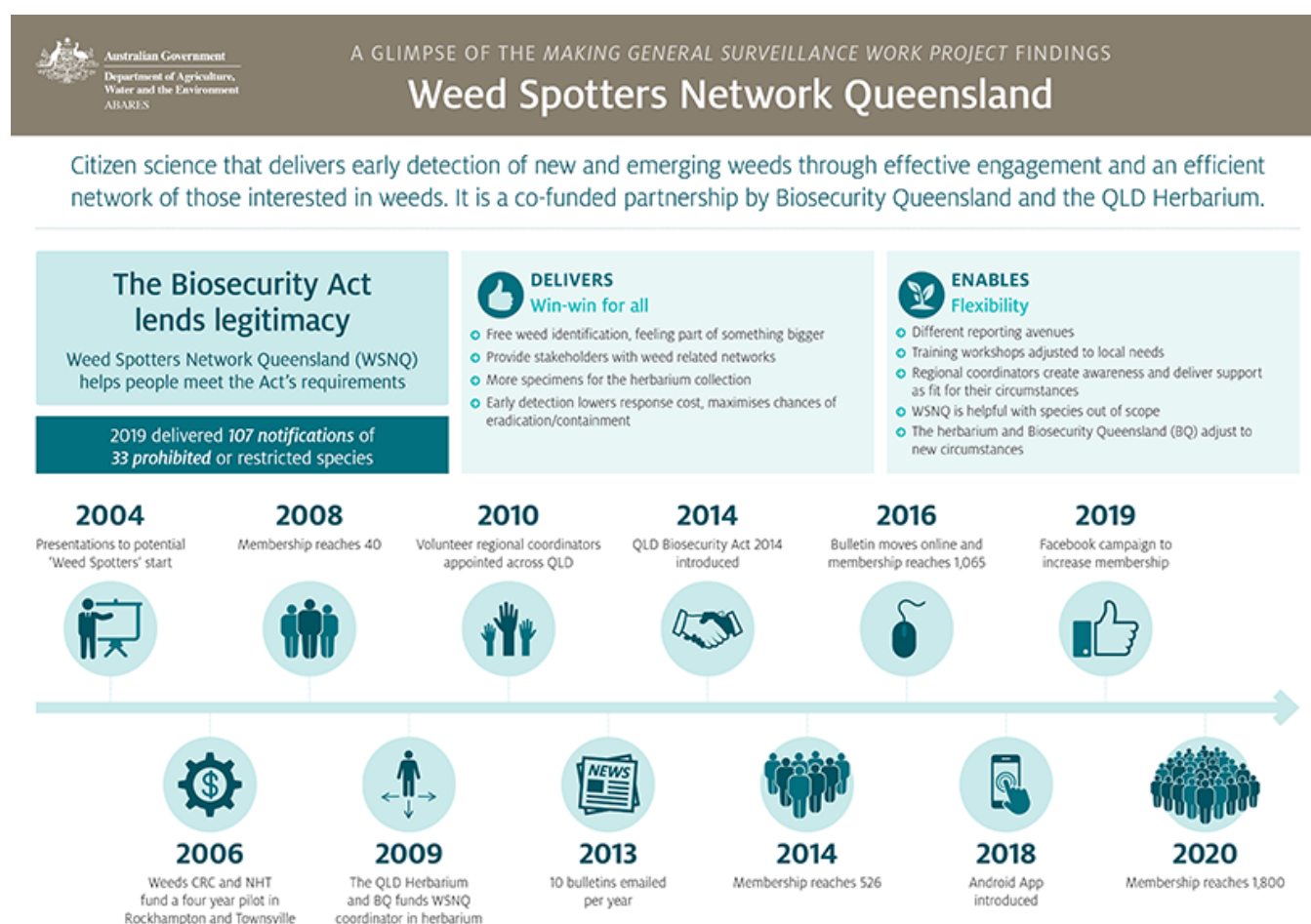
The case studies explored during Phase 2 of this project are listed below, including links to presentations that were prepared by the case study staff.

Note: As these documents have not been prepared by the department they may not meet [Australian Government accessibility guidelines](#). If you have difficulty accessing these files, visit [web accessibility](#) for assistance.

Scroll further down to find an infographic for each case study based on some of the project team's key findings.

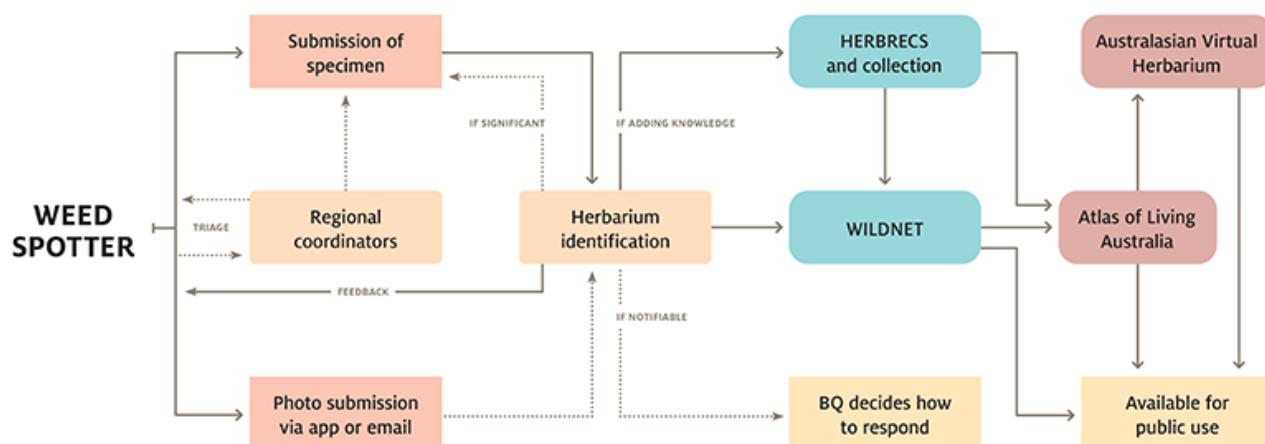
- [Weed Spotters Network Queensland](#)
- [Weed Spotters Victoria](#)
- [NZ General Surveillance Program – Plant health component](#)
- [FishWatch SA](#)
- [State Wide Array Surveillance Program \(SWASP\)](#)
- [MyPestGuide – Pantry Blitz](#)
- [Rural Practitioner Enhanced Disease Surveillance, South Australia](#)
- [Northern Australia Biosecurity Surveillance \(NABS\) Net](#)
- [Engaging Indigenous community about general surveillance](#)

Weed Spotters Network Queensland infographic



Weed Spotters Network Queensland

The data flow process



Spot-on spotting ...

In 2019 a local government officer attended a Weed Spotters workshop. Not long after the workshop he recognised some of the species he learned about in pots in someone's backyard. He reported it to Weed Spotters and a visit from BQ officers confirmed that it was Karroo thorn (*Vachellia karroo*). This exotic species has the potential to do more damage than the similar species, prickly acacia (*Vachellia nilotica*), one of Queensland's worst weeds, as it has a broader habitat tolerance and produces more seeds. They were swiftly destroyed.

Weed Spotters Network Victoria infographic



A GLIMPSE OF THE MAKING GENERAL SURVEILLANCE WORK PROJECT FINDINGS

Weed Spotters Victoria

Agriculture Victoria coordinates the targeted recruitment and training of people with the necessary skills, opportunity and motivation to report when they see any of a defined group of 8 to 12 of their State Prohibited Weeds.

2018 – Of 23 confirmed State Prohibited Weed reports, 39% were from Weed Spotters

Of 18 non-target weed reports 22% were from Weed Spotters



WEED SPOTTER CHARACTERISTICS

- Have a passion for biosecurity, their region and increasing their knowledge
- Enjoy free Weed Spotter training
- May be unsure of species identification but it's OK to submit a false report
- Don't want to feel implicated in the outcome of a report
- Like to feel connected and part of a larger program



MONITORING AND EVALUATION HAS LED TO A 'POLISHED' PROGRAM

- Engagement is adaptive and on-going (face-to-face, online newsletters, annual calendar, social media, online training)
- Recruitment is targeted to fill gaps in location and organisational roles
- Training is adapted, with a strong 'hands-on' live specimen component
- The number of focus weeds was reduced so to 'not dilute the message'



FLEXIBLE APPROACH TO MANAGE VARIABILITY ACROSS THE PROGRAM

- Variety in weed reports (i.e. land or water species, public or private land)
- Provide different reporting options (call centre, email, Regional and Local Biosecurity Officers)
- Different options for confirming ID (High Risk Invasive Plants (HRIP) staff, Regional and Local Biosecurity Officers, state herbariums)
- Because of the small number of reports, the process can be flexible
- Balance the number of Weed Spotters, their reports and the program resources

2001

Weed Spotters Victoria conceptualised



2005

Weed Alert Contact Officers (WACOs) appointed for each catchment



2006

Widespread training of Weed Spotters



2008

Strategic plan outlined targeted recruitment and training



2012

Improved Weed Spotter database and website



2020

Training became available online



2002

Weed Spotters begin to report weeds



2006

Weed Spotters is officially launched



2007

Database developed to support a coordinated approach



2008

Reduced target species to 8-12 State Prohibited Weeds

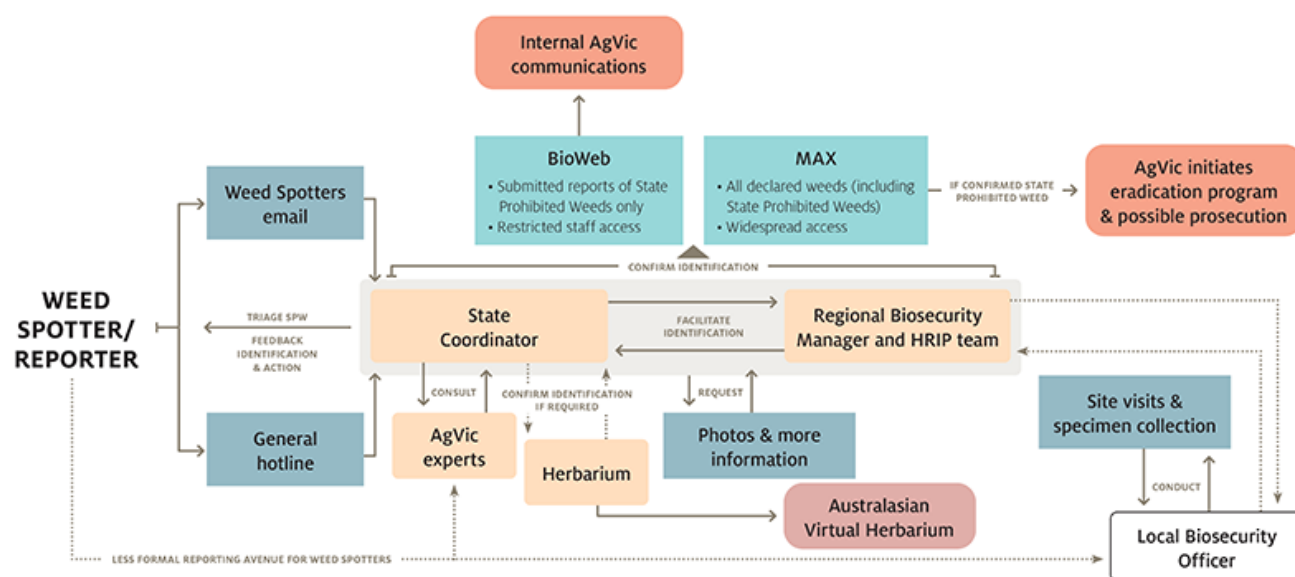


2015

Weed Spotter newsletter moved online

Weed Spotters Victoria

The data flow process



Spot-on spotting ...

Review of the geographical distribution of Weed Spotters in Victoria identified a gap in the Mildura region. Consequently, local training sessions were run to raise awareness of the focus State Prohibited Weeds at risk of incursion. Shortly after the training several accurate reports of State Prohibited Weeds were made, prompting new eradication programs in the area.

New Zealand General Surveillance Program – Plant health infographic



A GLIMPSE OF THE MAKING GENERAL SURVEILLANCE WORK PROJECT FINDINGS

New Zealand General Surveillance Program – Plant health

A hotline-based system that allows all New Zealanders to report suspected sightings of pests, weeds and diseases that is supported by targeted engagement of groups that have the motivation, capability, and access to report. It is funded by the NZ Ministry for Primary Industries (MPI).

2019 – the Plant Health team managed 1,323 notifications through general surveillance

Most (910) were from the public; of which 526 (58%) were investigated further



BALANCE INCOMING NOTIFICATIONS AND SPECIES IDENTIFICATION CAPACITY

- Triage by call centre and some by industry bodies
- Investing in community engagement to increase quality of incoming notifications
- Tapping into citizen science to increase reporting and lessen identification load through Find-A-Pest app
- Maximising lab capacity through experienced lab team and making case investigation a separate function
- On-going communication between communication teams and lab staff about planned campaigns



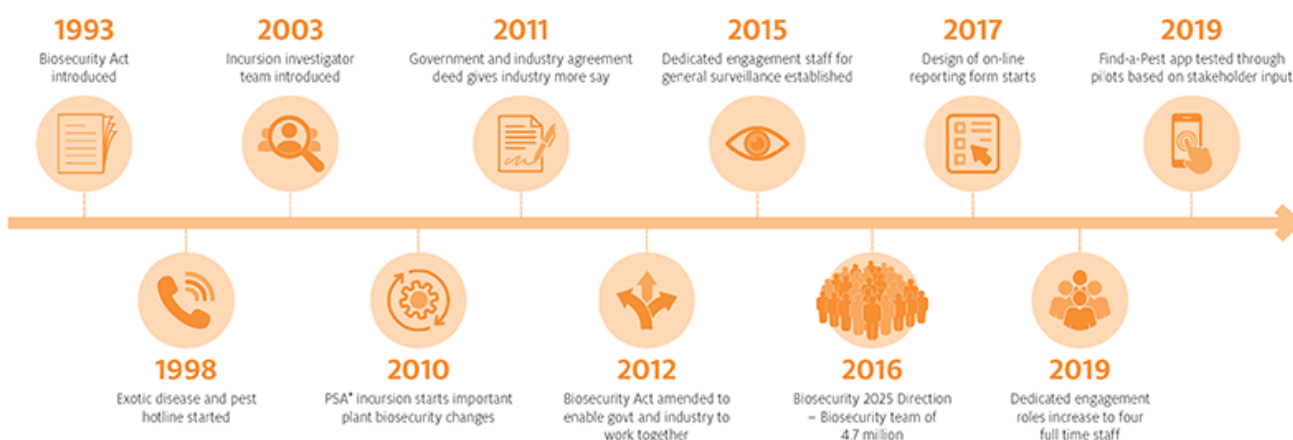
RESPONSIVE TO THE COMMUNITY'S EXPECTATIONS

- A mindset throughout the program
- Quick responses and feedback to hotline calls
- Evidence based campaigns and engagement
- Investing in staff to build relationships with key groups – set up to be agile, innovative, and able to respond to organic opportunities
- Introducing a range of reporting tools to meet different community needs



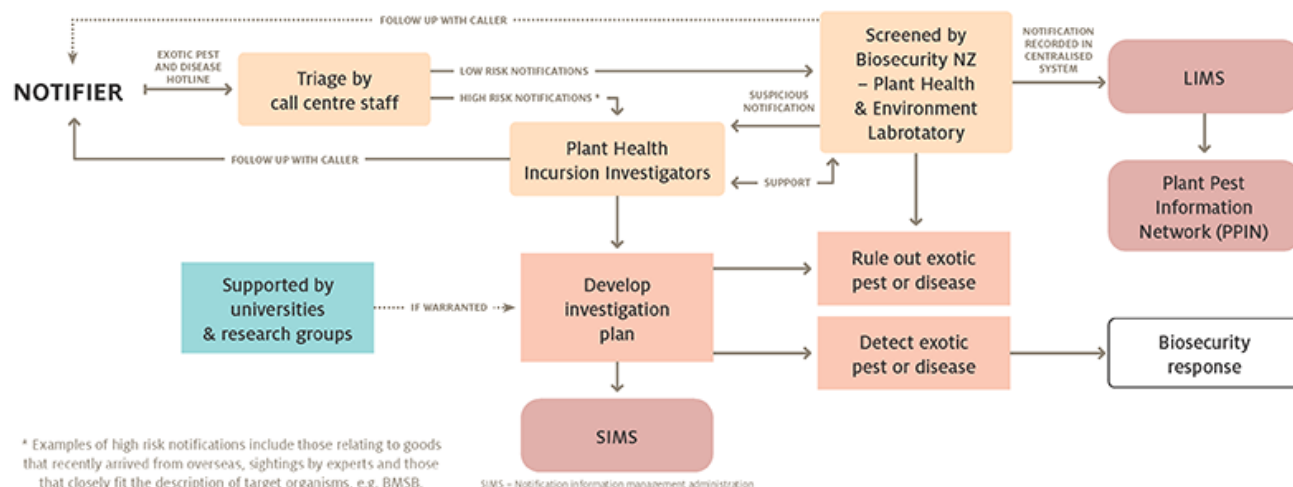
CONTINUALLY IMPROVE AND ADAPT

- Evaluate impact of campaigns; use findings to improve next round
- Use pilots to engage stakeholders, and test and refine new ideas, e.g. Find-A-Pest app
- Engage stakeholders and potential critics early on and bring them on the journey
- Effective communication channels throughout system, e.g. 'go to' people and working groups, so info flows timely to where needed



* *Pseudomonas syringae* pv. *actinidiae* – a serious bacterial kiwifruit vine disease

The data flow process



Spot-on spotting ...

A traveller found a suspected Brown Marmorated Stink Bug (BMSB) on a house floor while unpacking his luggage after he returned from a trip to Italy. He informed MPI about it via the Exotic Pest and Disease Hotline. The Plant Health and Environment Laboratory confirmed the identification from photos. On the request of an Incursion Investigator (II), the traveller searched and shaken all items in the suitcase and inspected the room for other stink bugs but none were found. An II visited the house to interview the notifier and inspect an unopened suitcase, but no more stink bugs were found.

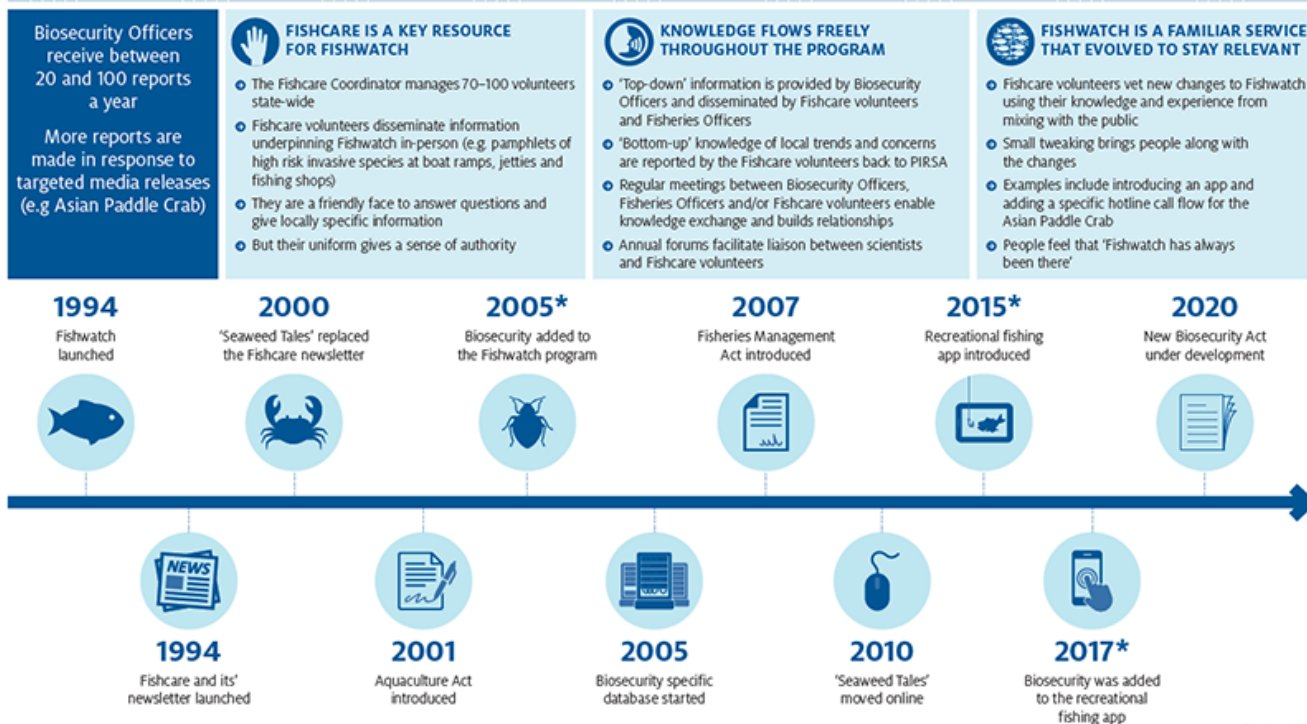
Fishwatch South Australia – Biosecurity infographic



A GLIMPSE OF THE MAKING GENERAL SURVEILLANCE WORK PROJECT FINDINGS

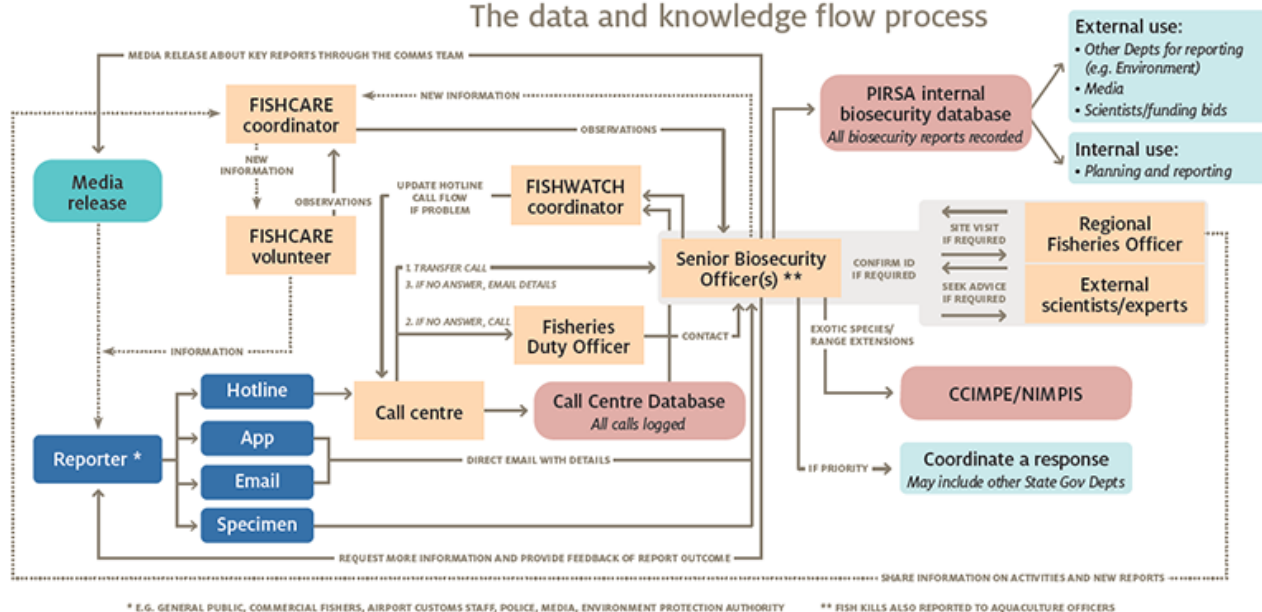
Fishwatch South Australia (Biosecurity Component)

Fishwatch provides a 'one-stop-shop' for the general public, commercial fishers and professionals (e.g. airport customs staff and police) to access information and report potential marine pest sightings or suspect fishing activities to PIRSA biosecurity, aquaculture and fisheries experts. This supports future marine and freshwater environments and their stock.



* Estimated time

The data and knowledge flow process



Spot-on spotting ...

In 2019 a commercial fisher found an Asian Paddle Crab in South Australian waters (www.nationaltribune.com.au). Upon receiving the report, PIRSA ran a targeted media campaign to increase awareness of the species and potential impact. This sparked a peak in biosecurity reports to the department.

State Wide Array Surveillance Program (SWASP) infographic



A GLIMPSE OF THE MAKING GENERAL SURVEILLANCE WORK PROJECT FINDINGS State-Wide Array Surveillance Program (SWASP)

SWASP has succeeded in providing robust, coordinated, collaborative and consistent surveillance of potential marine pests within ports across Western Australia (WA), filling an important gap in the existing marine biosecurity surveillance for the state.

Successes of the program:

- Collaborative
- Consistent across the state
- Effective in early detection



TRUST IS KEY TO SUCCESSFUL COLLABORATION

- There is strong trust between DPIRD staff and the ports
- It takes time and effort to build up trust
- Annual DPIRD visits to the ports helps to maintain the relationships and two-way communication
- Trust enables a strong collaboration between the regulator (DPIRD) and the ports and opens opportunities for collaboration on other issues
- Ports like to feel connected and part of a larger program



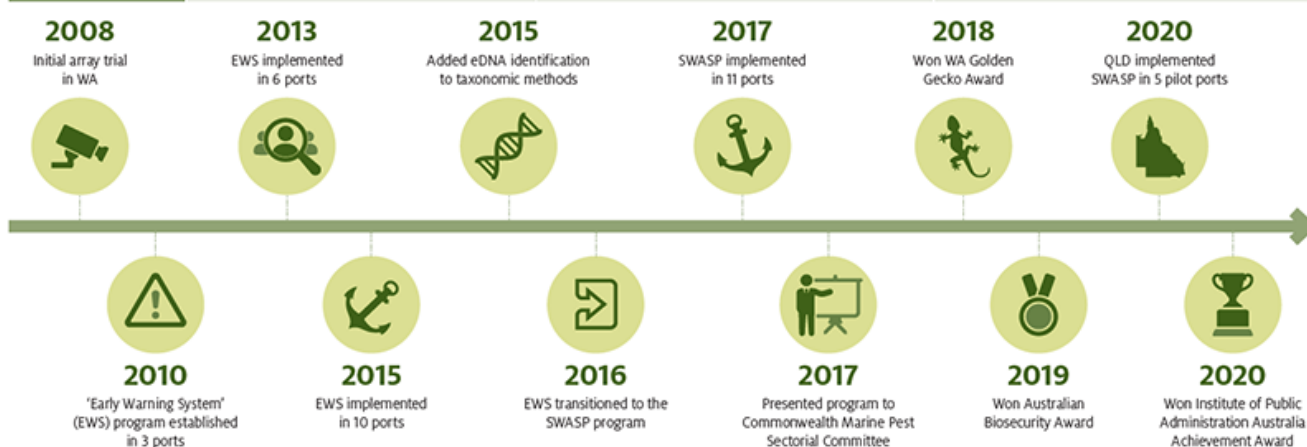
SWASP PROVIDES MORE THAN BIOSECURITY SURVEILLANCE

- Strong collaboration supports trialling new innovations
- Molecular ID enabled biodiversity data (not just species presence/absence) to be collected
- Ports utilise community interest in biodiversity information to increase engagement with them
- Community engagement increases general knowledge and awareness of invasive species

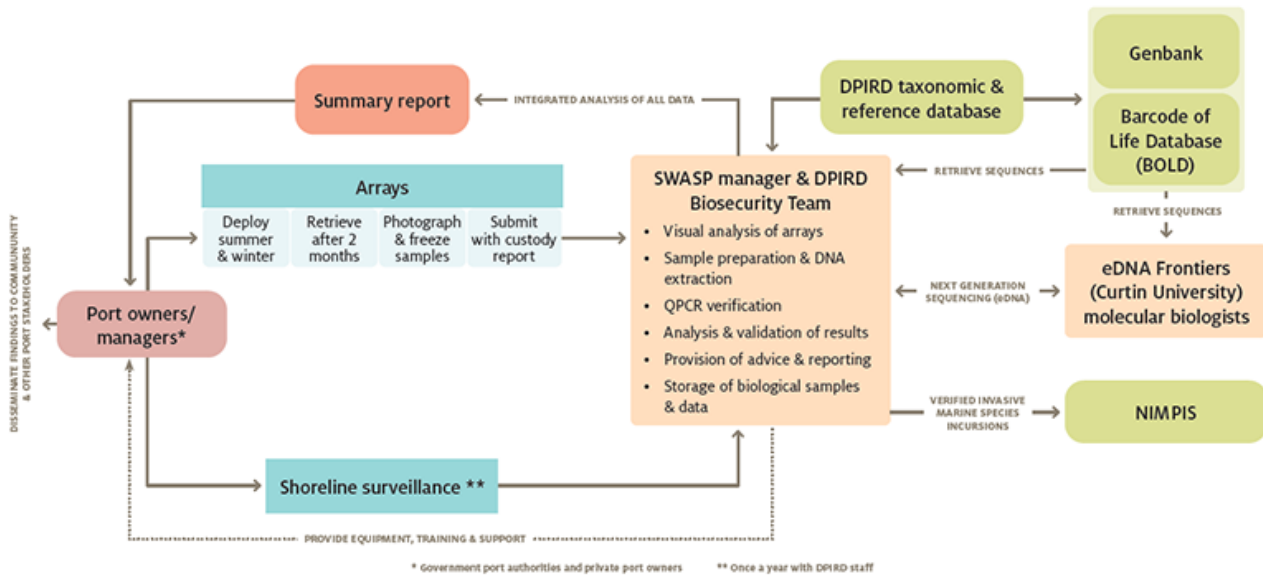


INNOVATION HAS ITS CHALLENGES

- Get the program up and running, then 'iron out the bugs' (e.g. Adapt surveillance design and equipment)
- Change is initiated by ports identifying a problem or DPIRD identifying an opportunity
- Change and innovation presents new challenges so SWASP needs to be adaptable
- Because the ports trust DPIRD they accept the challenge



The data flow process



Shining success ...

Since its establishment in just 3 ports, the SWASP program has collaboratively adapted, evolved, innovated and grown into an award-winning biosecurity surveillance program in 11 ports across the state. It won the WA Golden Gecko Award for environmental excellence (2018), Australian Biosecurity Award (2019) and an Institute of Public Administration Australia Achievement Award (2020). However, the creation of a strong network of government regulators, port authorities and the general public who are now aware and involved in protecting WA from invasive marine pests, is valued as a greater achievement.

MyPestGuide Pantry Blitz infographic



A GLIMPSE OF THE MAKING GENERAL SURVEILLANCE WORK PROJECT FINDINGS

The WA Pantry Blitz

Members of the public place sticky traps with a combined Khapra beetle lure in their pantries. They make weekly reports for one month by submitting photos through the MyPestGuide™ Reporter app (MPG-RA). The Department of Primary Industries and Regional Development, Western Australia (DPIRD) developed the app. The data collected can provide supporting evidence of pest freedom if trading partners enquire about the status of Khapra beetle in WA.

DPIRD received 2252 reports in 2016 and over 2600 reports in 2017

Data quality is maintained by using a combined Khapra beetle lure and the appropriate skilled triage officer and entomologist confirming each report



CONSIDERABLE INTERNAL LIAISON AND ADJUSTMENT NEEDED

- Faced initial scepticism
- Limited existing guidance, e.g. privacy matters
- Needed to establish clear roles and responsibilities across various areas in DPIRD
- Needed to establish clear pathways, e.g. for samples to reach entomologist via post
- Most effective community engagement communication style did not match existing Department's communication style
- The need for urgency was not always understood



EFFECTIVE APP DEVELOPMENT REQUIRES MUCH EFFORT, WIDE INPUT AND ON-GOING UPDATES

- Wide engagement of scientists, taxonomists, statisticians, community, etc. is needed
- Must be intuitive and simple for users, no more than 4 clicks
- Consider back-end design, e.g. how data will be used, best format, avoid double data handling
- Test with various users and mobile phones
- Keep updated, have prototypes for testing, including for app updates
- Account for privacy concerns, e.g. can switch GPS off



IDENTIFICATION CAPACITY CRUCIAL

- Vital to maintaining data quality
- Additional load on triage officer and entomologists is easily underestimated or overlooked
- Identification staff usually very busy, cannot rely on their goodwill only
- Quick ID response to participants crucial to maintain their engagement
- ID staff may need to learn to interact with community
- May need to pay for ID staff's service

2000

Pantry Survey – participants sent trap back, 15% response



2009/10

The department faces resource cuts, incl. to surveillance



2014

The department develops MPG-RA for the community



2016

First Pantry Blitz – first Blitz that 'ticks all targeted surveillance boxes'



2017

Second Pantry Blitz with lessons incorporated from first Pantry Blitz



2021

Third Pantry Blitz in March



2007

Biosecurity and Agriculture Management Act facilitates community involvement in biosecurity



2014

The department develops new pest ID app for industry



2015

First Biosecurity Blitz



2016/17

Updates to MPG-RA based on lessons learned

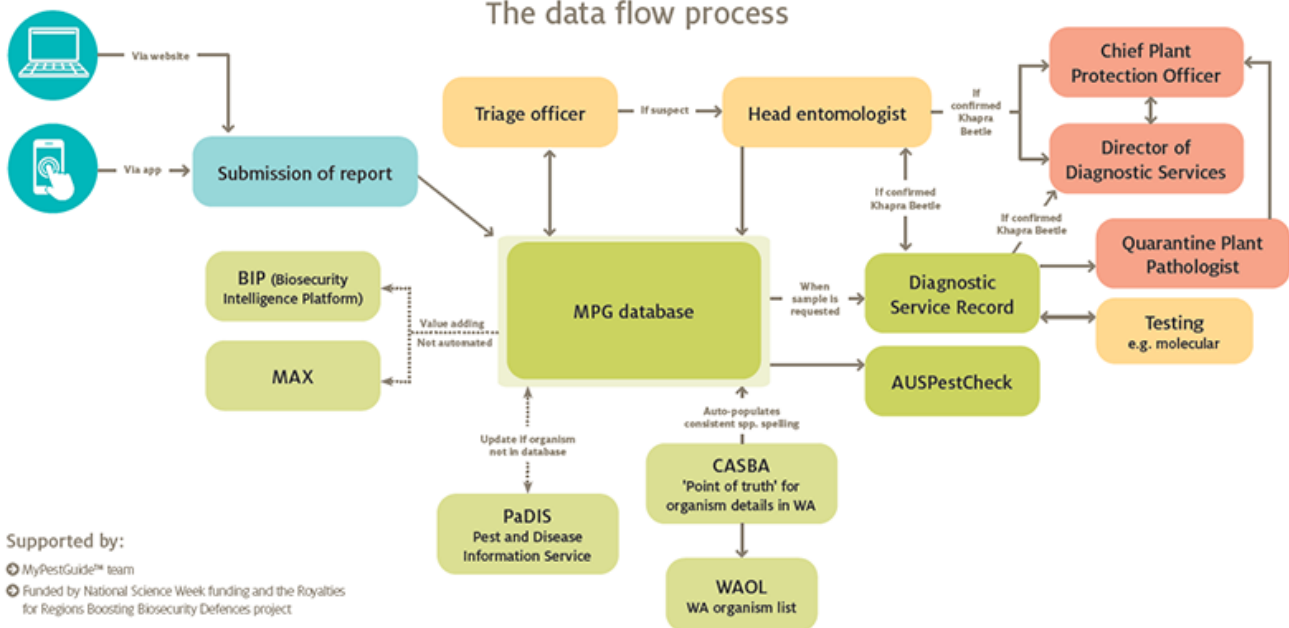


2019

Plant Health Committee agreed to have MPG-RA as the preferred general surveillance app across Australia

The WA Pantry Blitz

The data flow process



Supported by:

- MyPestGuide™ team
- Funded by National Science Week funding and the Royalties for Regions Boosting Biosecurity Defences project

Larvae in your larder?

The 2016 Pantry Blitz activity was a great success with 1877 participants registering for a free trap and the department receiving 2252 biosecurity surveillance reports. Communications reached over 250,000 people and the campaign won the WA Rural Media Best Campaign award for 2016. The department uses the submitted pest reports to support Western Australia's pest free status from Khapra beetle, which ensures WA's ongoing access to international markets.

Northern Australia Biosecurity Surveillance Network (NABSnet)

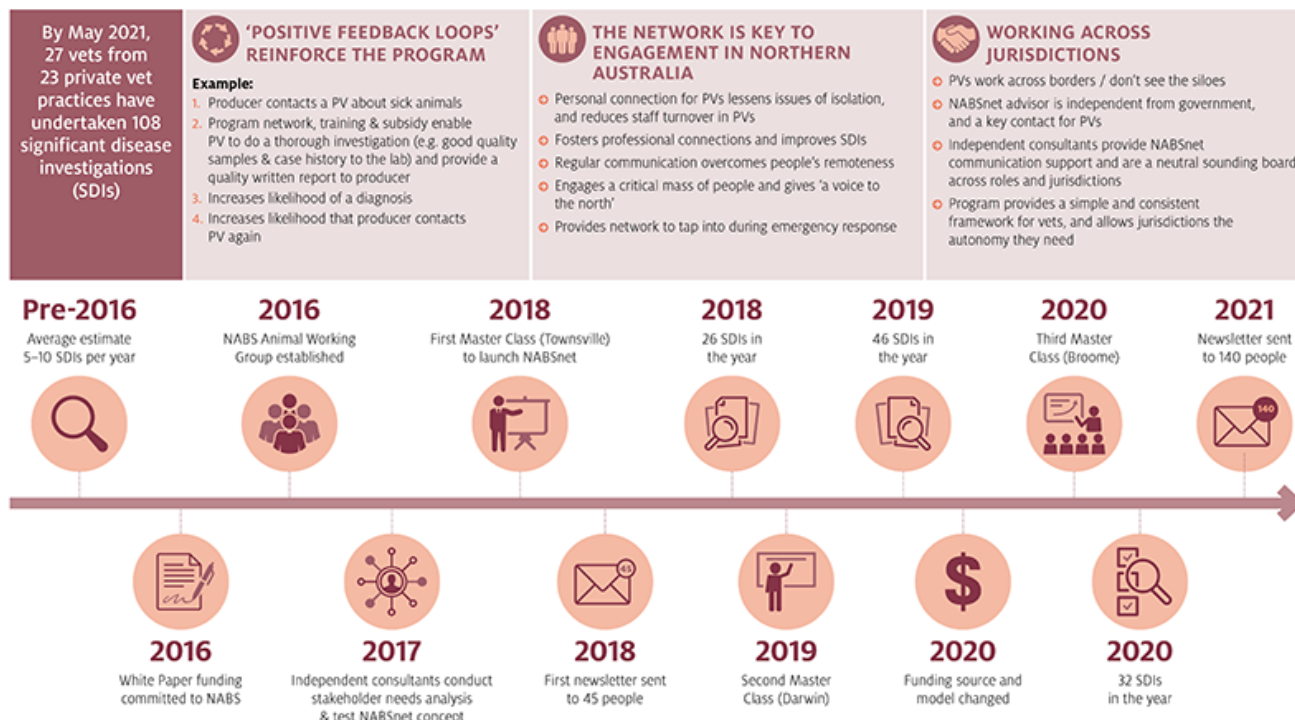
infographic



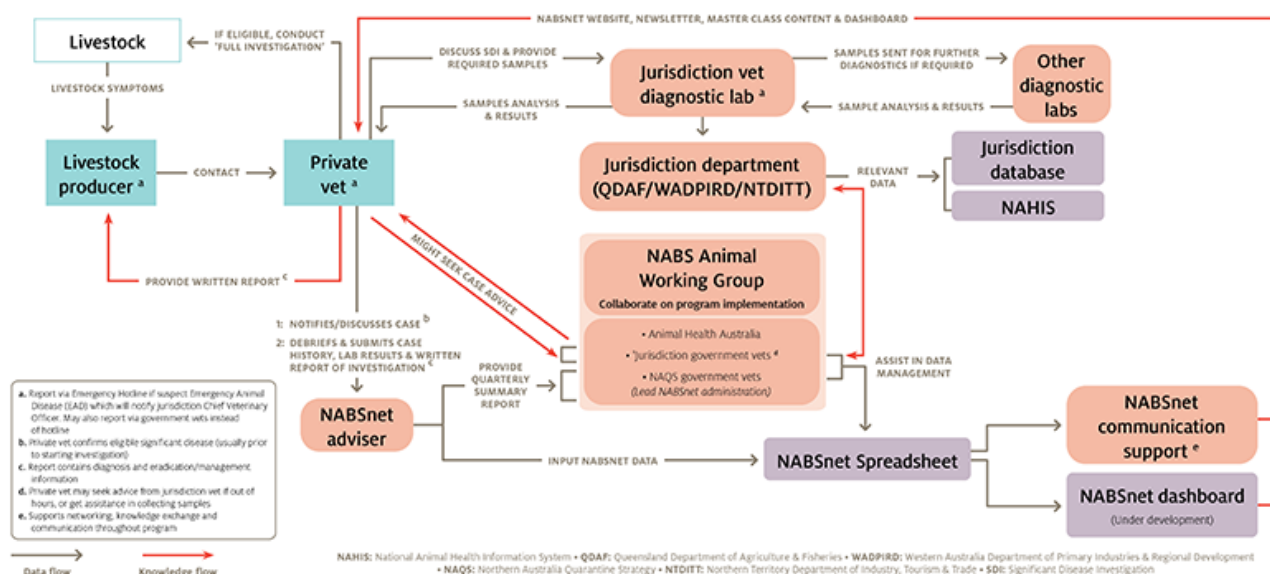
A GLIMPSE OF THE MAKING GENERAL SURVEILLANCE WORK PROJECT FINDINGS

Northern Australia Biosecurity Surveillance Network (NABSnet)

NABSnet utilises the coverage, expertise and good will of private vets (PVs) working in northern Australia to improve general surveillance for animal pests and diseases. In return it provides PVs access to practical training and resources to improve their capability to do quality significant disease investigations (SDIs); a subsidy for conducting SDIs; and networking opportunities to develop relationships between PVs across northern Australia and people in the government biosecurity sector (particularly vets and laboratory staff).



The data and knowledge flow process



Master class underpins collaboration throughout the network ...

The Master Class provides private vets and lab pathologists with the opportunity to socialise, build two-way understanding and foster working friendships. Consequently, they are able to self-organise to better meet each other's needs. Writing thorough case notes is difficult in a private vet's busy schedule, while pathologists need good case information to decide which tests to undertake. Some private vets and pathologists are now comfortable with the vet calling the pathologist while driving to or from a job, to provide the case information verbally, and the pathologist types this information into the lab's database.

SA Rural Practitioner Enhanced Disease Surveillance Program

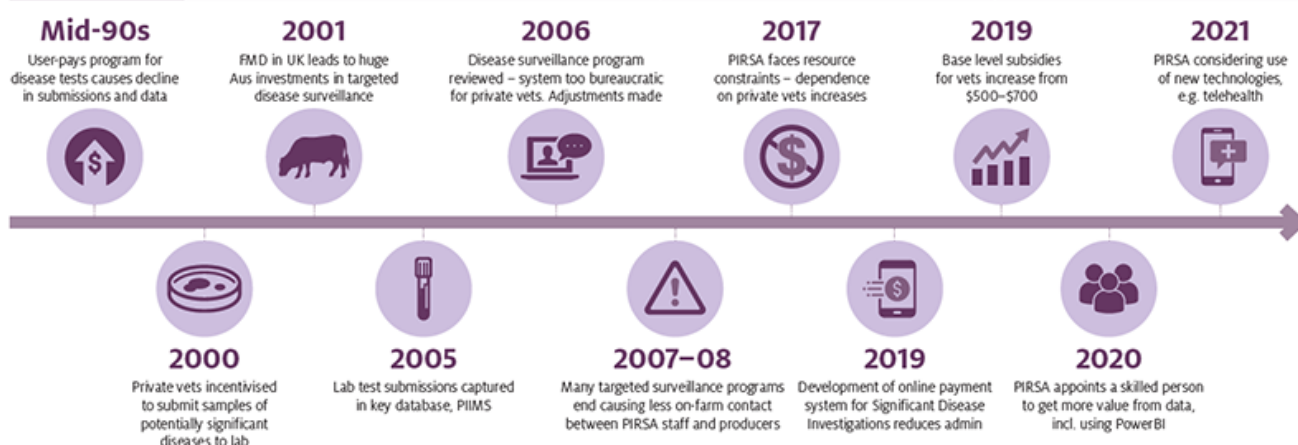


A GLIMPSE OF THE MAKING GENERAL SURVEILLANCE WORK PROJECT FINDINGS

SA Rural Practitioner Enhanced Disease Surveillance Program

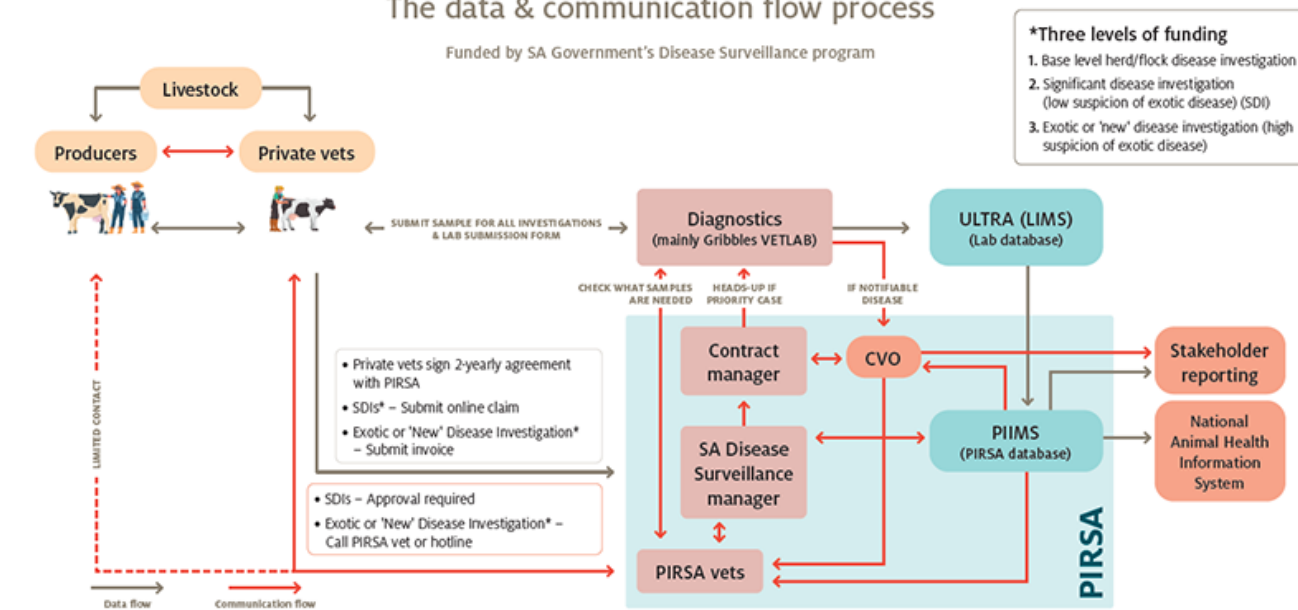
Primary Industries and Regions South Australia (PIRSA) provides subsidies for private veterinary investigations into livestock diseases involving laboratory tests to rule out notifiable diseases and where an infectious agent is a potential cause. Five PIRSA vets are appointed to oversee certain livestock species and certain regions. They build trust relationships with private vets to encourage and support them in their participation. The Rural Practitioner Enhanced Disease Surveillance Program (RPEDSP) assists PIRSA to provide proof of freedom from certain diseases to international markets and to detect exotic or newly emerging diseases as early as possible.

Between 30–36 private vet practices varying in size, support the program	KNOWLEDGE BROKERING AND NETWORKS ARE VITAL - PIRSA vets assist time poor private vets with RPEDSP requirements and interpretation of some lab reports. They also relay other important info, e.g. emergency response arrangements - PIRSA vets have valuable species- and region-specific knowledge - PIRSA vets assist with skill transfer e.g. demonstration of taking unusual samples - Vets highly value networks with other vets for support when unusual or complex cases occur. Events with other vets are key to building contacts - Some specialised PIRSA staff help lab staff see the context of the samples they receive, e.g. prevalent diseases in an industry	NEED TO WORK WITH PRIVATE VETS' EXPECTATIONS - Can't focus on collecting disease rule-outs only, the program must benefit private vets, e.g. assist them to provide a better service to their clients - Heavy admin requirements deter vets' participation or cause a need for PIRSA staff to follow-up - PIRSA vets ensure support for private vets is a phone call away	DATA SYSTEM MUST MANAGE WEAK POINTS - Clearly defined and communicated roles and responsibilities across data flow pathways are key - Capable people across data system are vital to ensure vulnerabilities are minimised and data value is maximised - Avoid key functions in data management resting with one person
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The data & communication flow process

Funded by SA Government's Disease Surveillance program



Shining success ...

In 2021, a cattle producer noticed a cow with a nasal discharge, with sores on her nose, ulcers in her mouth and cloudy eyes. He called his local private vet, who was part of the Rural Practitioner Enhanced Disease Surveillance program. One of the differential diagnoses was foot and mouth disease (FMD), so the vet took a detailed history and submitted the appropriate samples to the laboratory. PIRSA funded the investigation, which resulted in FMD being ruled out, and the endemic disease, malignant catarrhal fever to be confirmed. This information is important to prove to interstate and international trading partners that SA (and Australia) remains free of FMD, and to support early detection of potential emergency animal diseases.

Toolburn, J. & Cooper, H. 2021. A glimpse of the 'making general surveillance work' project findings: SA Rural Practitioner Enhanced Disease Surveillance Program, ABARES, Bioscience Infographic 21.5, Canberra, Wap. DOI: <https://doi.org/10.25816/infog-845>. CC BY 4.0.

Source: <https://www.agriculture.gov.au/abares/research-topics/social-sciences/making-general-surveillance-work/general-surveillance-case-study-overviews>